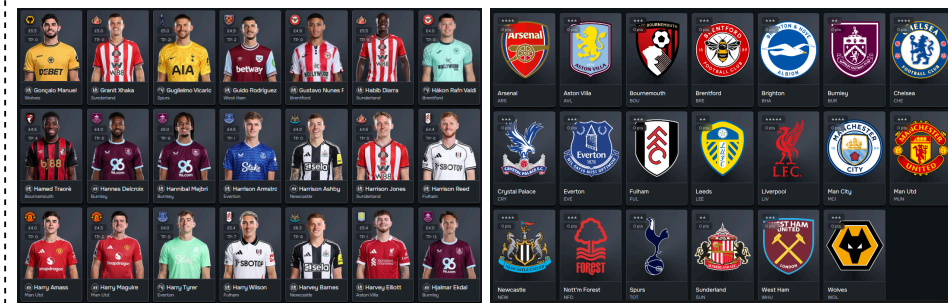


Candidate player–gameweek data



- Player, club, position, and gameweek
- Price
- Historical FPL points, minutes, and starts
- xG, xA, xGI, xGC, ICT, and selection percentage
- Availability/missing-gameweek information

Fixed-squad policy evaluation and reported outputs: GWs 27–38

- Construct a new 15-player squad once before GW27 using GWs 1–26
- Compare static and fixed-squad sequential lineup policies
- Sequential policy changes only the starting XI and bench order
- Squad membership, captain, and vice-captain remain fixed
- Formation-valid automatic substitutions
- Zero external transfers and transfer-hit deductions; no chips are used
- Calculate weekly and cumulative realized points
- Regularization, budget, Pareto, and availability sensitivity

Data preprocessing and leakage control

- Validate schema, coerce data types, deduplicate, and sort by player and GW
- Training period: GWs 1–26; evaluation period: GWs 27–38
- Treat missing player–GW observations as unavailable
- Count missing observations as zero starts for the availability estimator
- Use only lagged or pre-deadline information
- Exclude target-GW points, minutes, bonus, and BPS
- Use price only in optimization constraints, not as a predictor

Training-period diagnostics and feature specifications

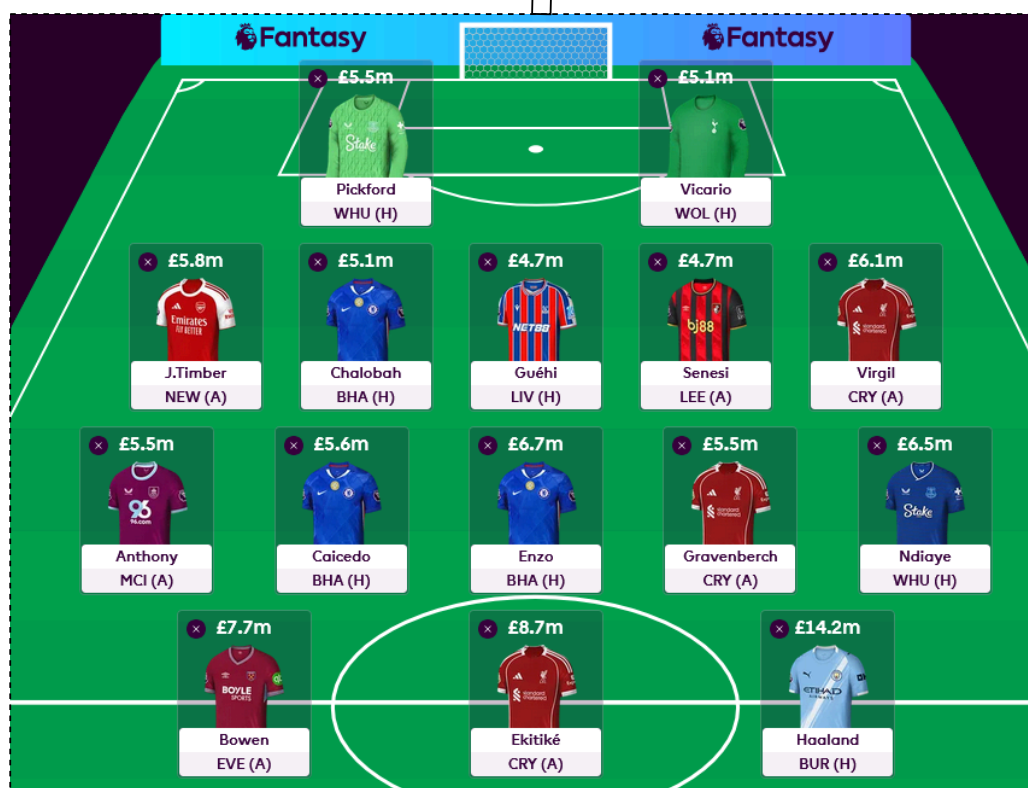
- Position-specific correlations and feature distributions
- Redundancy audit: $xGI \approx xG + xA$
- XGBoost/SHAP diagnostics for interpretation only
- Full, Reduced, and Parsimonious Ridge specifications
- Full LASSO for automatic feature selection
- XGBoost/SHAP are used only for interpretation and not supplied to the MILP

Forecasting and cost-vector construction

1. Simple and recency-weighted averages; Holt smoothing
2. Linear trend and ARIMA variants
3. Nonparametric bootstrap and parametric Monte Carlo simulation
4. Position-specific Ridge/LASSO and hybrid recent-points–Ridge models
5. Alternative cost vectors: ICT, robust ICT, and $xGI - xGC$
6. Box-robust and availability-adjusted coefficients

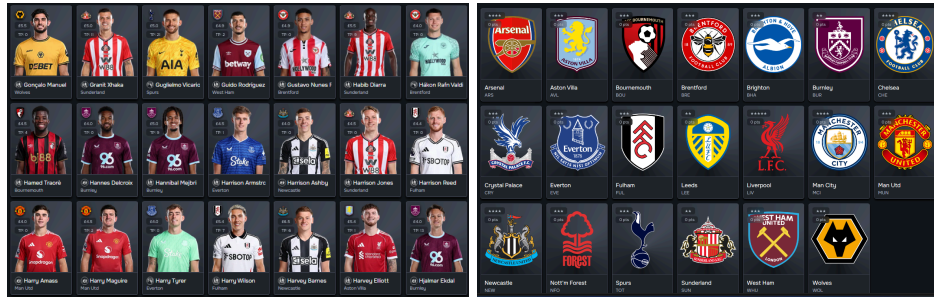
Roster optimization

- **Two-stage baselines**
 - Heuristic fixed allocation: £83.5m XI and £16.5m bench
 - Actual-remainder rule: unused XI allowance passes to the bench
- **Joint 15-player MILP**
 - Select squad, starting XI, and captain simultaneously
 - Total budget £100m
- **Constraints**
 - Squad quotas: 2–5–5–3
 - Legal starting formation
 - At most three players per club
 - Captain belongs to starting XI
- **Epsilon-constraint Pareto analysis**
 - Trade-off between predicted XI-plus-captain value F_{XI} and bench value F_B



Illustrative 15-player squad visualization—not an experimental result

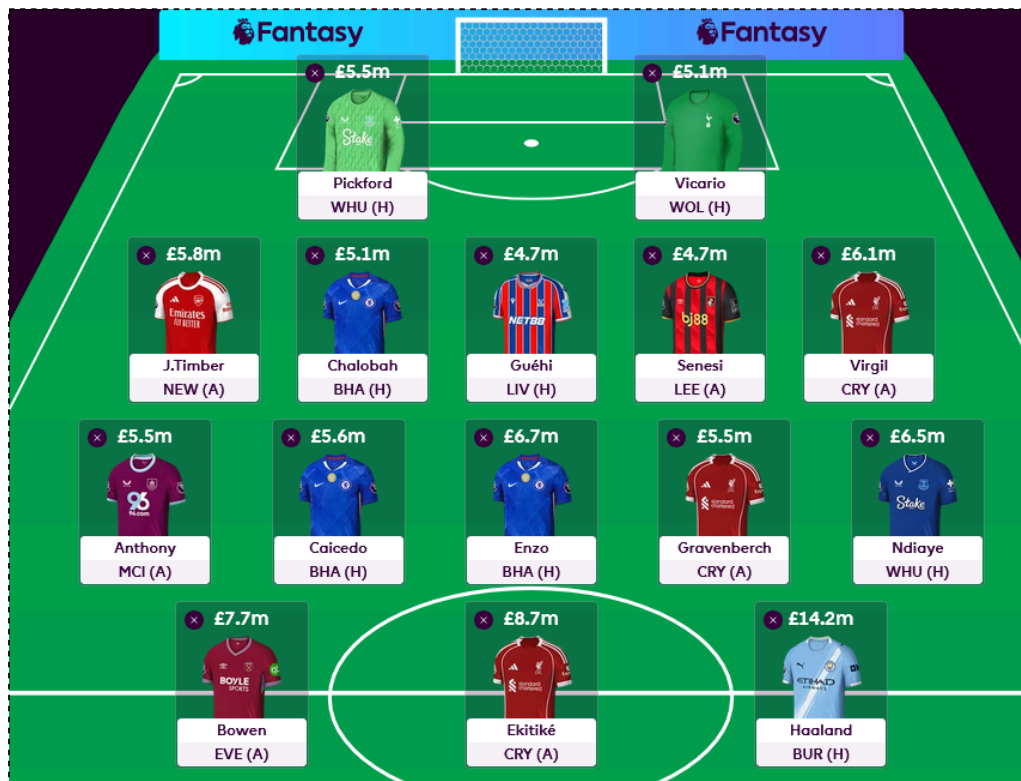
List of players from different teams



- Player list (name, club, position)
- Price
- Historical points
- Minutes/starts

Outputs & evaluation

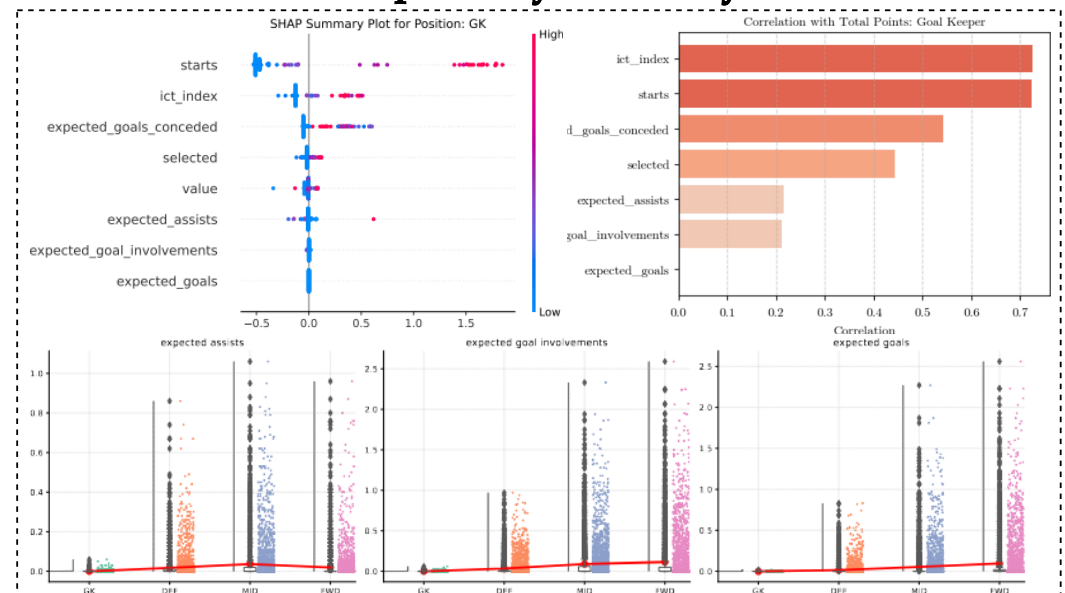
- Recommended XI, captain, bench; formation trends
- Weekly & cumulative points; method leaderboard
- Budget sensitivity; rolling-window effects
- Diagnostics/interpretation



Preprocessing and feature selection

- Reshape/verify player-GW panel; coerce types & dedupe
- Build table schema for target GW: name, position, team, value, slot for expected points (filled by models later)
- Sort by GW per player; treat missing weeks = unavailable
- Train/Test: 1-26 → 27-38; guard against leakage (no same-GW points/bonus/BPS as inputs)
- Simple/weighted avg, ARIMA, Monte Carlo: time-series on total_points
- Hybrid score: ict_index, xG, xA, xGI, xGC, selected, starts
- Surrogate objectives in the XI optimizer: xGI - xGC, ICTGI - xGC, ICT

Exploratory data analysis



Modeling (expected points)

1. Averages/smoothing
2. ARIMA variants
3. Linear trend regression
4. Simulation: bootstrap & parametric
5. Hybrid: ridge + recent points.

Optimization (team selection)

- 0-1 Integer Programming: choose XI + captain
- Constraints: budget, formation, ≤ 3 /club, captain $\in$ XI
- Bench optimizer (remaining budget, fills to 2-5-5-3)
- Robust variant (penalize score by uncertainty)